



| IN                   |  | anat_template_wf | OUT                         |  |                       |
|----------------------|--|------------------|-----------------------------|--|-----------------------|
| inputnode.anat_files |  |                  | outputnode.anat_realign_xfm |  | outputnode.out_report |
|                      |  |                  | outputnode.anat_ref         |  |                       |
|                      |  |                  | outputnode.anat_valid_list  |  |                       |
|                      |  |                  | outputnode.out_report       |  |                       |

| inputnode |  | OUT          |  |  |  |  |
|-----------|--|--------------|--|--|--|--|
| IN        |  | flair        |  |  |  |  |
|           |  | roi          |  |  |  |  |
|           |  | subject_id   |  |  |  |  |
|           |  | subjects_dir |  |  |  |  |
|           |  | t1w          |  |  |  |  |
|           |  | t2w          |  |  |  |  |

|              |              |     |
|--------------|--------------|-----|
| IN           | fs_isrunning | OUT |
| subject_id   | utility      | out |
| subjects_dir |              |     |

|         |               |          |
|---------|---------------|----------|
| IN      | anat_validate | OUT      |
|         | header        |          |
| in_file |               | out_file |

| IN                 |  | brain_extraction_wf | OUT                       |  |                     |                     |                     |
|--------------------|--|---------------------|---------------------------|--|---------------------|---------------------|---------------------|
| inputnode.in_files |  |                     | outputnode.bias_corrected |  | outputnode.out_file | outputnode.out_mask | outputnode.out_segm |
|                    |  |                     | outputnode.out_file       |  |                     |                     |                     |
|                    |  |                     | outputnode.out_mask       |  |                     |                     |                     |
|                    |  |                     | outputnode.out_segm       |  |                     |                     |                     |
|                    |  |                     | outputnode.out_segm       |  |                     |                     |                     |

| IN          |  | t1w_buffer | OUT         |  |
|-------------|--|------------|-------------|--|
| ants_seg    |  |            | ants_seg    |  |
| t1w_brain   |  | utility    | t1w_brain   |  |
| t1w_mask    |  |            | t1w_preproc |  |
| t1w_preproc |  |            |             |  |

|              |                   |              |
|--------------|-------------------|--------------|
| IN           | sourcefile_buffer | OUT          |
| source_files | utility           | source_files |

| IN                   |  | t2w_template_wf | OUT                 |  |
|----------------------|--|-----------------|---------------------|--|
| inputnode.anat_files |  |                 | outputnode.anat_ref |  |

| IN                         |  | surface_recon_wf | OUT                         |                       |
|----------------------------|--|------------------|-----------------------------|-----------------------|
| inputnode.flair            |  |                  | outputnode.fsnative2t1w_xfm | outputnode.subject_id |
| inputnode.skullstripped_t1 |  |                  |                             |                       |
| inputnode.subject_id       |  |                  |                             |                       |
| inputnode.subjects_dir     |  |                  |                             |                       |
| inputnode.t1w              |  |                  |                             |                       |
| inputnode.t2w              |  |                  |                             |                       |

| IN                          |  | ds_fs_registration_wf | OUT                          |  |
|-----------------------------|--|-----------------------|------------------------------|--|
| inputnode.fsnative2anat_xfm |  |                       | outputnode.fsnative2anat_xfm |  |
| inputnode.source_files      |  |                       |                              |  |

| IN           |            | bbreg | OUT          |
|--------------|------------|-------|--------------|
| source_file  | freesurfer |       | out_ita_file |
| subject_id   |            |       |              |
| subjects_dir |            |       |              |

| IN                     | gifti_spheres_wf |    |
|------------------------|------------------|----|
| inputnode.subject_id   |                  |    |
| inputnode.subjects_dir |                  | ou |

| IN                     | gifti_morphometrics_wf | OUT                  |
|------------------------|------------------------|----------------------|
| inputnode.subject_id   |                        | outputnode.sulc      |
| inputnode.subjects_dir |                        | outputnode.thickness |

| IN                     |  | ds_spheres_wf |
|------------------------|--|---------------|
| inputnode.source_files |  |               |
| inputnode.sphere       |  |               |

| IN                     |  | ds_morph_wf |   |
|------------------------|--|-------------|---|
| inputnode.source_files |  |             | o |
| inputnode.sulc         |  |             |   |

|                   |                |           |
|-------------------|----------------|-----------|
| IN                | fs2anat_buffer | OUT       |
|                   | utility        |           |
| fsnative2anat_xfm |                | fsnative2 |

| IN                                                                                |                             | refinement_wf | OUT                      |
|-----------------------------------------------------------------------------------|-----------------------------|---------------|--------------------------|
|  | inputnode.ants_segs         |               | outputnode.out_brainmask |
|                                                                                   | inputnode.fsnative2anat_xfm |               |                          |
|                                                                                   | inputnode.reference_image   |               |                          |
|                                                                                   | inputnode.subject_id        |               |                          |
|                                                                                   | inputnode.subjects_dir      |               |                          |

|     |            |     |
|-----|------------|-----|
| IN  | merge_xfms | OUT |
| in1 | utility    | out |
| in2 |            |     |

| IN                          |  | gifti_surface_wf | OUT                     |  |
|-----------------------------|--|------------------|-------------------------|--|
| inputnode.fsnative2anat_xfm |  |                  | outputnode.midthickness |  |
| inputnode.subject_id        |  |                  | outputnode.pial         |  |
| inputnode.subjects_dir      |  |                  | outputnode.white        |  |

| IN                     |  | ds_surfaces_wf | OUT                     |  |
|------------------------|--|----------------|-------------------------|--|
| inputnode.midthickness |  |                | outputnode.midthickness |  |
| inputnode.pial         |  |                | outputnode.pial         |  |
| inputnode.source_files |  |                | outputnode.pial         |  |
| inputnode.white        |  |                | outputnode.white        |  |

| IN           |  | surfaces_buffer | OUT          |  |
|--------------|--|-----------------|--------------|--|
| midthickness |  |                 | midthickness |  |
| pial         |  | utility         | pial         |  |
| sphere       |  |                 | sphere       |  |
| sphere_reg   |  |                 | sphere_reg   |  |
| sulc         |  |                 | sulc         |  |
| thickness    |  |                 | thickness    |  |
| white        |  |                 | white        |  |

| IN                     | cortex_masks_wf | OUT                     |
|------------------------|-----------------|-------------------------|
| inputnode.midthickness |                 | outputnode.cortex_masks |
| inputnode.thickness    |                 | outputnode.source_files |

| IN                     |  | ds_cortex_masks_wf | OUT                   |  |
|------------------------|--|--------------------|-----------------------|--|
| inputnode.mask_files   |  |                    | outputnode.mask_files |  |
| inputnode.source_files |  |                    |                       |  |

| IN          | ds_t2w_preproc | OUT      |
|-------------|----------------|----------|
| in_file     |                | out_file |
| source_file |                |          |

| IN                     |  | ds_t1w_mask_wf | OUT                  |
|------------------------|--|----------------|----------------------|
| inputnode.mask_file    |  |                | outputnode.mask_file |
| inputnode.source_files |  |                |                      |

| IN                     |  | register_template_wf | OUT                     |
|------------------------|--|----------------------|-------------------------|
| inputnode.lesion_mask  |  |                      | outputnode.anat2std_xfm |
| inputnode.moving_image |  |                      | outputnode.std2anat_xfm |
| inputnode.moving_mask  |  |                      | outputnode.template     |

| IN                 |  | anat_ribbon_w |
|--------------------|--|---------------|
| inputnode.pial     |  |               |
| inputnode.ref_file |  |               |
| inputnode.white    |  |               |

| IN       | fast | OUT                  |
|----------|------|----------------------|
|          | fsl  | partial_volume_files |
| in_files |      | partial_volume_map   |

| IN                   | fsLR_reg_wf | OUT                        |
|----------------------|-------------|----------------------------|
| inputnode.sphere_reg |             | outputnode.sphere_reg_fsLR |
|                      |             |                            |

|         |              |     |        |           |
|---------|--------------|-----|--------|-----------|
| IN      | lut_t1w_dseg | OUT | IN     | fast2bids |
|         | utility      |     |        | utility   |
| in_dseg |              | out | inlist |           |

| IN                        |  | ds_fsLR_reg_wf | OUT                        |  |
|---------------------------|--|----------------|----------------------------|--|
| inputnode.source_files    |  |                | outputnode.sphere_reg_fsLR |  |
| inputnode.sphere_reg_fsLR |  |                |                            |  |

|        |           |     |
|--------|-----------|-----|
| IN     | fast2bids | OUT |
| inlist | utility   | out |
|        |           |     |

| IN                        |  | ds_fsLR_reg_wf | OUT                        |
|---------------------------|--|----------------|----------------------------|
| inputnode.source_files    |  |                | outputnode.sphere_reg_fsLR |
| inputnode.sphere_reg_fsLR |  |                |                            |

|           |                        |            |                      |
|-----------|------------------------|------------|----------------------|
|           | IN                     | ds_tpms_wf | OUT                  |
| anat_dseg | inputnode.anat_tpms    |            | outputnode.anat_tpms |
|           | inputnode.source_files |            |                      |

| IN                     | ds_tpms_wf | OUT                  |
|------------------------|------------|----------------------|
| inputnode.anat_tpms    |            | outputnode.anat_tpms |
| inputnode.source_files |            |                      |

| IN              | fsLR_buffer | OUT             |
|-----------------|-------------|-----------------|
| sphere_reg_fsLR | utility     | sphere_reg_fsLR |
|                 |             |                 |

|     |                 |     |
|-----|-----------------|-----|
| IN  | anat2std_buffer | OUT |
|     | utility         |     |
| in2 |                 | out |

| IN                        |  | msm_sulc_wf | OUT                        |
|---------------------------|--|-------------|----------------------------|
| inputnode.sphere          |  |             |                            |
| inputnode.sphere_reg_fsLR |  |             | outputnode.sphere_reg_fsLR |

| IN       | seg_buffer | OUT      |
|----------|------------|----------|
| t1w_dseg | utility    | t1w_dseg |
| t1w_tpms |            | t1w_tpms |